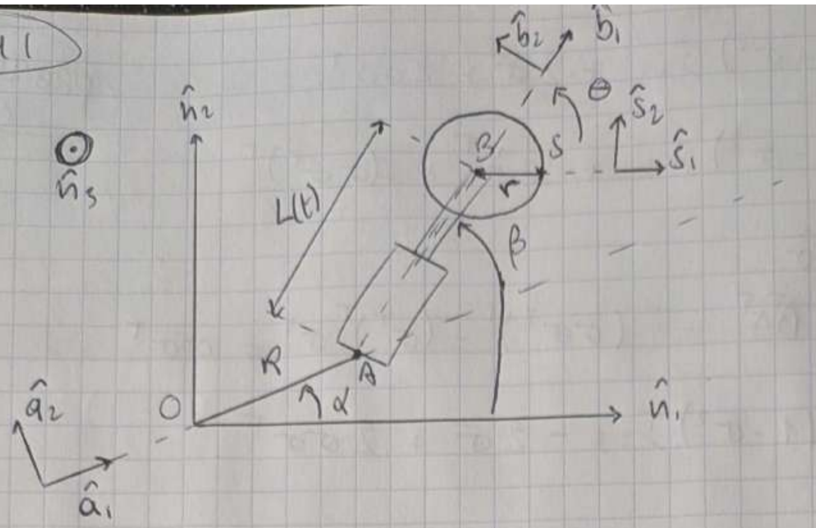
PROBLEM 1



Rotating frames

$A: \{\hat a_1, \hat a_2, \hat n_3\}$

$B: \{\hat b_1, \hat b_2, \hat n_3\}$

$S: \{\hat s_1, \hat s_2, \hat n_3\}$

$N: \{\hat n_1, \hat n_2, \hat n_3\}$

α and β are inertialy defined wrt O :

$$\omega_{A/O} = \dot\alpha\, \hat n_3$$

$$\omega_{B/O} = \cancel{\omega_{B/A} + \omega_{A/O}} = (\cancel{\dot\alpha} + \dot\beta)\,\hat n_3 = \underline{\dot\beta\, \hat n_3}$$

The disk that S is attached to rotates with constant spin vector relative to the piston :

$$\omega_{S/B} = \dot\theta\, \hat n_3 \quad \text{and} \quad \ddot\theta = 0$$

$$\omega_{S/O} = \omega_{S/B} + \omega_{B/O} + \cancel{\omega_{A/O}} = (\cancel{\dot\alpha} + \dot\beta + \dot\theta)\,\hat n_3$$

The position of S relative to the inertial frame can be written as :

$$\rho_{S/O} = \rho_{S/B} + \rho_{B/A} + \rho_{A/O}$$

$$\rho_{S/O} = \rho_S = R\hat a_1 + L(t)\hat b_1 + r\hat s_1$$

(1) Inertial velocity of point S : $\dot\rho_S = \frac{{}^N d}{dt}(\rho_S)$

$$\dot\rho_S = \frac{{}^N d}{dt}(R\hat a_1) + \frac{{}^N d}{dt}(L(t)\hat b_1) + \frac{{}^N d}{dt}(r\hat s_1)$$

Transport theorem for a vector $\vec u$ wrt a body frame B :

$$\frac{{}^N d}{dt}(\vec u) = \frac{{}^B d}{dt}(\vec u) + (\omega_{B/N} \times \vec u)$$

R and r being constant scalars : $\dot R = \dot r = 0$.

Applying this to the previous equation :

- $${}^{N}\frac{d}{dt}(\hat{a}_1) = \underbrace{{}^{A}\frac{d}{dt}(\hat{a}_1)}_{0} + (\omega_{A/0} \times \hat{a}_1) \quad , \quad \omega_{A/0} = \dot{\alpha}\,\hat{n}_3$$

$${}^{N}\frac{d}{dt}(\hat{a}_1) = \dot{\alpha}\,\hat{a}_2$$

- $${}^{N}\frac{d}{dt}(\hat{b}_1) = \underbrace{{}^{B}\frac{d}{dt}(\hat{b}_1)}_{0} + (\omega_{B/0} \times \hat{b}_1) \quad , \quad \omega_{B/0} = \dot{\beta}\,\hat{n}_3$$

$${}^{N}\frac{d}{dt}(\hat{b}_1) = \dot{\beta}\,\hat{b}_2$$

- $${}^{N}\frac{d}{dt}(\hat{s}_1) = \underbrace{{}^{S}\frac{d}{dt}(\hat{s}_1)}_{0} + (\omega_{S/0} \times \hat{s}_1) \quad , \quad \omega_{S/0} = (\dot{\beta} + \dot{\theta})\,\hat{n}_3$$

$${}^{N}\frac{d}{dt}(\hat{s}_1) = (\dot{\beta} + \dot{\theta})\,\hat{s}_2$$

Then :

$$\boxed{\dot{\rho}_s = R\dot{\alpha}\,\hat{a}_2 + \dot{L}\,\hat{b}_1 + L\dot{\beta}\,\hat{b}_2 + r(\dot{\beta} + \dot{\theta})\,\hat{s}_2}$$

② Inertial acceleration of S : $\ddot{\rho}_S = {}^{N}\frac{d}{dt}(\dot{\rho}_S)$

$$\ddot{\rho}_S = R\,{}^{N}\frac{d}{dt}(\dot{\alpha}\,\hat{a}_2) + {}^{N}\frac{d}{dt}(\dot{L}\,\hat{b}_1) + {}^{N}\frac{d}{dt}(L\dot{\beta}\,\hat{b}_2) + r\,{}^{N}\frac{d}{dt}((\dot{\beta} + \dot{\theta})\,\hat{s}_2)$$

Using the transport theorem in a similar way

$$\ddot{\rho}_S = R\ddot{\alpha}\,\hat{a}_2 + R(\omega_{A/0} \times \hat{a}_2) + \ddot{L}\,\hat{b}_1 + \dot{L}(\omega_{B/0} \times \hat{b}_1)$$
$$+ \dot{L}\dot{\beta}\,\hat{b}_2 + L\ddot{\beta}\,\hat{b}_2 + L\dot{\beta}(\omega_{B/0} \times \hat{b}_2)$$
$$+ r(\ddot{\beta} + \ddot{\theta})\,\hat{s}_2 + r(\dot{\beta} + \dot{\theta})(\omega_{S/0} \times \hat{s}_2)$$

We said earlier that $\ddot{\theta} = 0$, we can then write :

$$\ddot{\rho}_S = -R\dot{\alpha}\,\hat{a}_1 + R\ddot{\alpha}\,\hat{a}_2 + \ddot{L}\,\hat{b}_1 + \dot{L}\dot{\beta}\,\hat{b}_2 + \dot{L}\dot{\beta}\,\hat{b}_2$$
$$+ L\ddot{\beta}\,\hat{b}_2 - L\dot{\beta}^2\,\hat{b}_1 + r(\ddot{\beta} + 0)\,\hat{s}_2 - r(\dot{\beta} + \dot{\theta})^2\,\hat{s}_1$$

$$\boxed{\begin{aligned}\ddot{\rho}_S = & -R\dot{\alpha}\,\hat{a}_1 + R\ddot{\alpha}\,\hat{a}_2 + (\ddot{L} - L\dot{\beta}^2)\,\hat{b}_1 + (L\ddot{\beta} + 2\dot{L}\dot{\beta})\,\hat{b}_2 \\ & - r(\dot{\beta} + \dot{\theta})^2\,\hat{s}_1 + r\ddot{\beta}\,\hat{s}_2\end{aligned}}$$

③ The position of A relative to S is :

$$\rho_{A/S} = \rho_{A/B} + \rho_{B/S} = -L(t)\,\hat{b}_1 - r\,\hat{s}_1$$

The corresponding velocity is then derived :

$$\dot{\rho}_{A/S} = \frac{{}^{S}d}{dt}(-L\hat{b}_1) + \underbrace{\frac{{}^{S}d}{dt}(-r\hat{s}_1)}_{0}$$

$$\dot{\rho}_{A/S} = -\dot{L}\,\hat{b}_1 - L\,(\omega_{B/S} \times \hat{b}_1) \quad , \quad \omega_{B/S} = -\omega_{S/B} = -\dot{\theta}\,\hat{n}_3$$

$$\dot{\rho}_{A/S} = -\dot{L}\,\hat{b}_1 - L(-\dot{\theta}\,\hat{b}_2)$$

$$\boxed{\dot{\rho}_{A/S} = -\dot{L}\,\hat{b}_1 + L\dot{\theta}\,\hat{b}_2}$$